Genome Forge Training Tutorial

Purpose: hands-on training for core Genome Forge workflows.

Audience: new users with zero prior context.

Start app: python3 web_ui.py --port 8080
Open: http://127.0.0.1:8080

This PDF mirrors docs/tutorial/user_training_tutorial.html.

Step 1: Load Sequence and Inspect Basics

1) Paste FASTA/GenBank/EMBL/raw DNA in the sequence panel.
2) Set name and topology.
3) Click Refresh Info.
4) Confirm length, GC topology, and feature count.

Tip: EMBL input is supported with feature parsing.

Step 2: Render Plasmid Map + Restriction Cuts

1) Use enzymes (for example EcoRI,BamHI).
2) Click Render Map.
3) Use wheel zoom + drag pan.
4) Click features/cuts to inspect details.

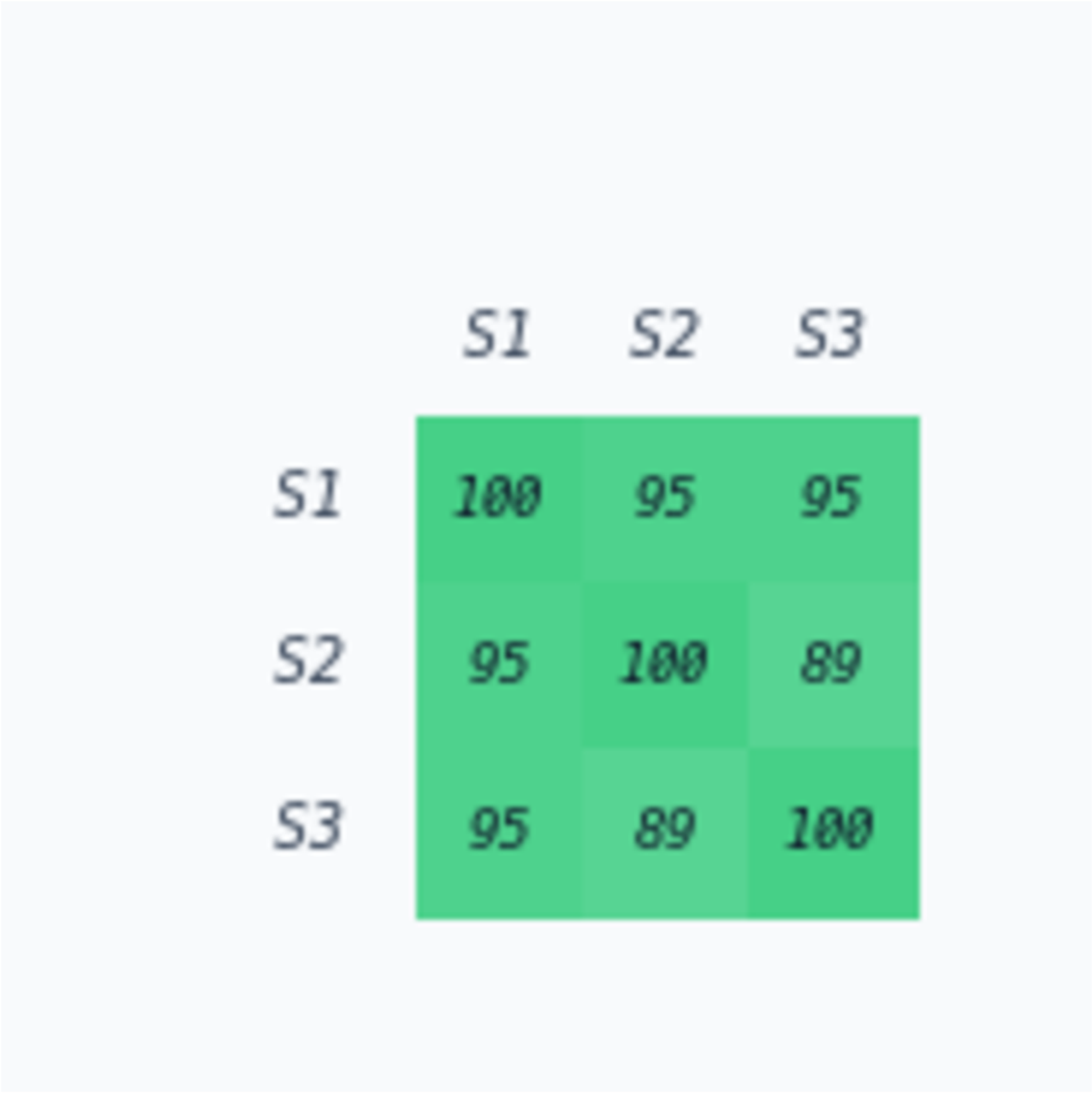
Goal: verify annotation placement and cut-site strategy.

Step 3: Sequence Track + Translation

1) Set track start/end and frame.
2) Render Sequence Track.
3) Use minimap brush and handles to navigate long regions.
4) Click codons to inspect residue mapping and frame behavior.

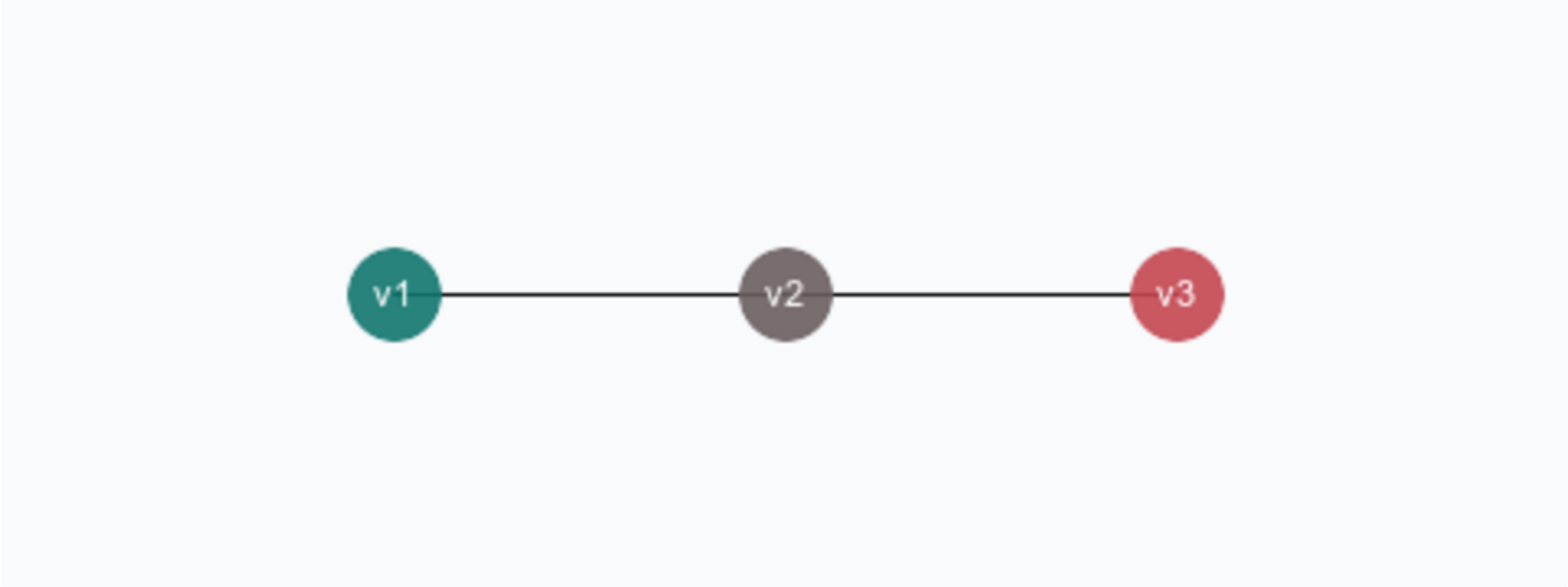
Step 4: Alignment and Identity Heatmap

1) Enter multiple sequences in Advanced tab.
2) Run MSA (progressive or external adapter).
3) Render identity heatmap and consensus.
4) Use this to verify expected variant similarity.



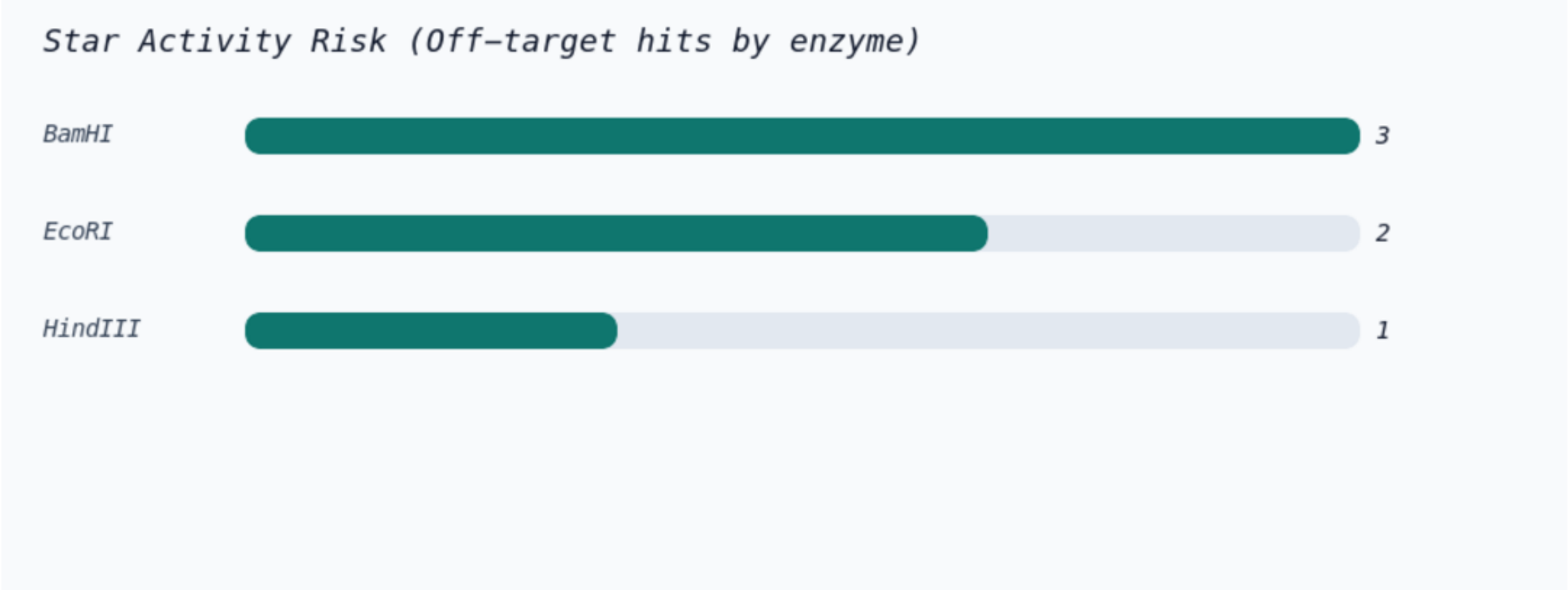
Step 5: History and Change Tracking

1) Save project snapshots.
2) Render Project History Graph.
3) Read recency coloring to find latest versions quickly.
4) Use undo/redo for local edit iteration (persisted in browser).



Step 6: Star Activity Risk Screening

1) Configure enzymes and star activity level.
2) Run Star Activity Scan.
3) Review off-target burden per enzyme.
4) Click top-risk cuts to center the sequence track.



Step 7: Ligation Product Ranking

1) Enter vector/insert sequences and enzyme ends.
2) Simulate ligation products.
3) Review desired vs byproduct classes and probabilities.
4) Inspect junction integrity and frame risk details.

Steps 8-9: Primer/PCR/Gel + Project Sharing

8) Design primers, run PCR, then simulate gel lanes with marker ladders.
9) Save projects, organize collections, create share bundles, and use /share/<id> pages for handoff.

Recommended daily loop:
Map -> Track -> Primers/PCR -> Cloning checks -> Save/Share